

WASTE OF SPACE - MARS TRASH EJECTOR

HALF HOLLOW HILLS EAST • MR. REGINI • HHHWASTEOFSPACE.WEBLY.COM

SOPHIA TRAN • ANYAA SACHDEV • MOHAMMAD WASIM • IAN SELIZHUK

ENGINEERING GOALS

1 SIMPLICITY

Simple, reliable mechanism with minimal points of failure.

2 MODULARITY

Crew can troubleshoot and replace components independently.

3 SAFETY

Multiple failsafes prevent accidental depressurization or ejection.

4 EFFICIENCY

Low power draw and fast cycle time to minimize crew downtime.

MATERIAL SELECTION

COMPONENT	MATERIAL	WHY
Chassis	Aluminum 6061	Lightweight, space-proven
Hatches	Titanium	Handles extreme temps
Seals	Silicone Rubber	Airtight in vacuum
Gears & Rack	Stainless Steel	Durable under stress
Electronics	Rad-Hard PCB	Resists Mars radiation
Fasteners	Titanium Bolts	Corrosion resistant
Ejection Plate	Carbon Fiber	Lightweight, rigid

EJECTION DATA

TRIAL	VERTICAL HEIGHT (M)	TIME IN AIR (S)	HORIZONTAL DISTANCE (M)	HORIZONTAL VELOCITY (M/S)
1	0.66	0.37	0.32	0.87
2	0.66	0.37	0.34	0.93
3	0.66	0.37	0.33	0.90
4	0.66	0.37	0.35	0.95
5	0.66	0.37	0.31	0.84
6	0.66	0.37	0.34	0.93
7	0.66	0.37	0.33	0.90
8	0.66	0.37	0.34	0.93
AVG	0.66	0.37	0.33	0.91

ELECTRONICS

SERVO

- Continuous rotation
- Open 560ms / Close 610ms

LIMIT SWITCH

- Detects inner hatch closed
- Checked every 10ms

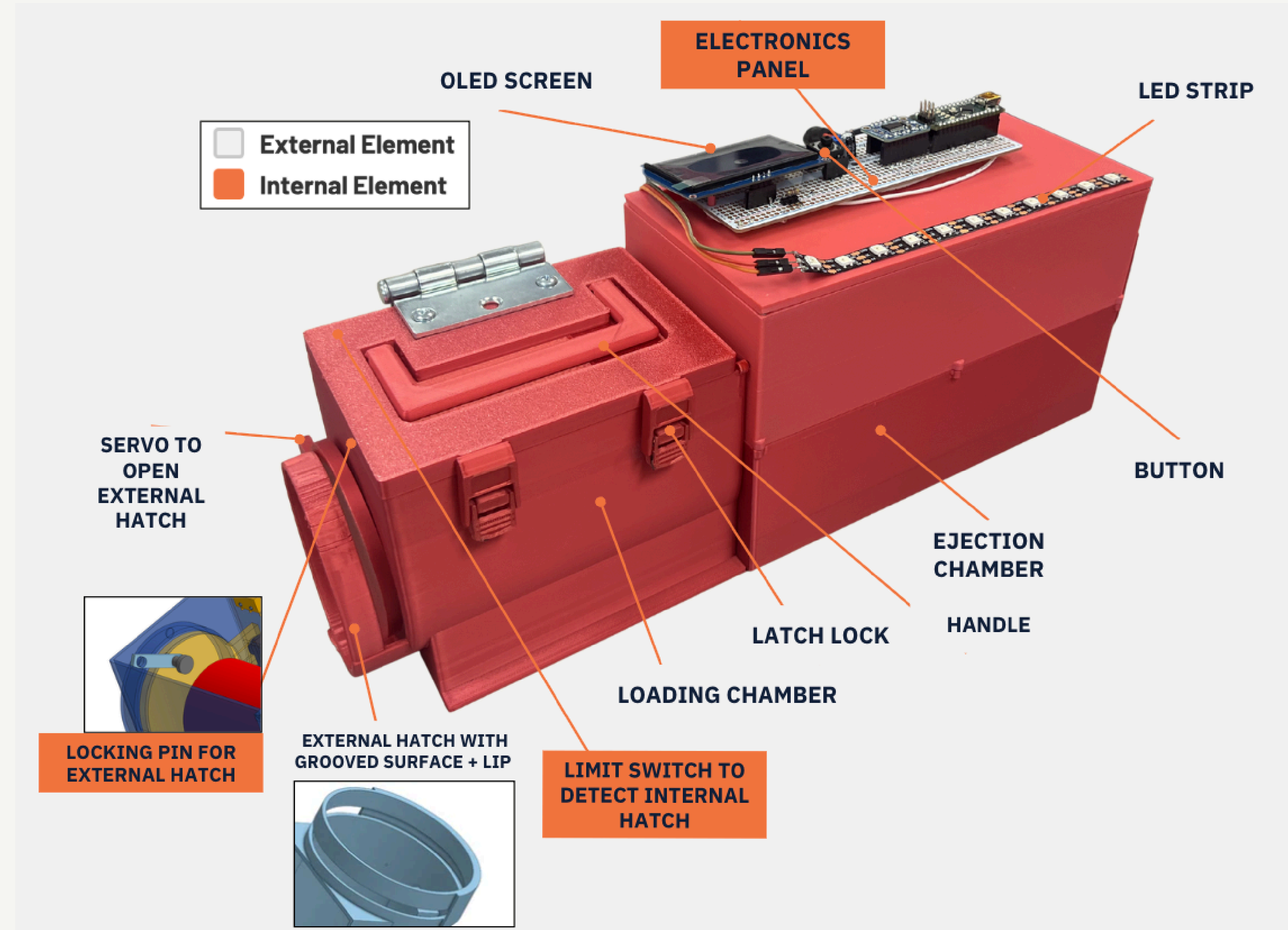
NEOPIXEL STRIP

- 6 RGB LEDs, status colors

OLED DISPLAY

- 128x32, I2C
- Phase names & errors

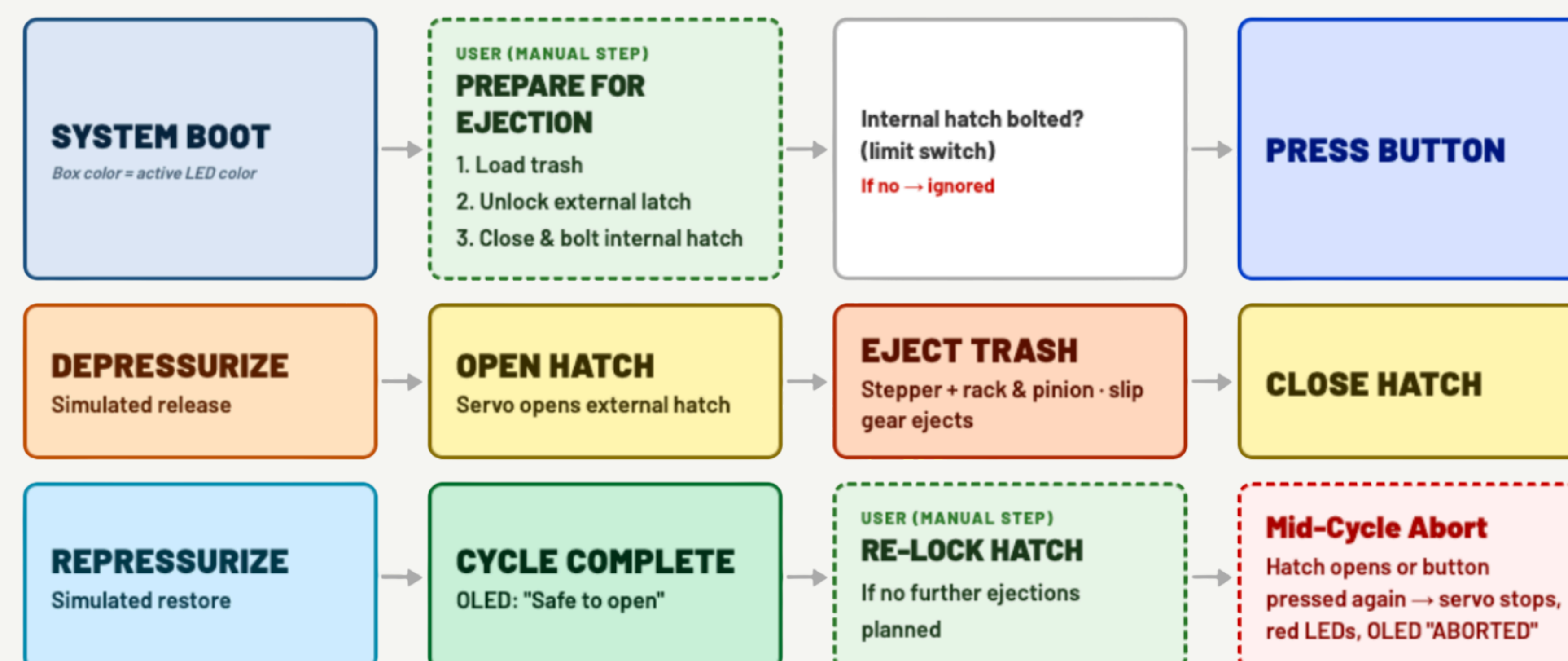
PROTOTYPE



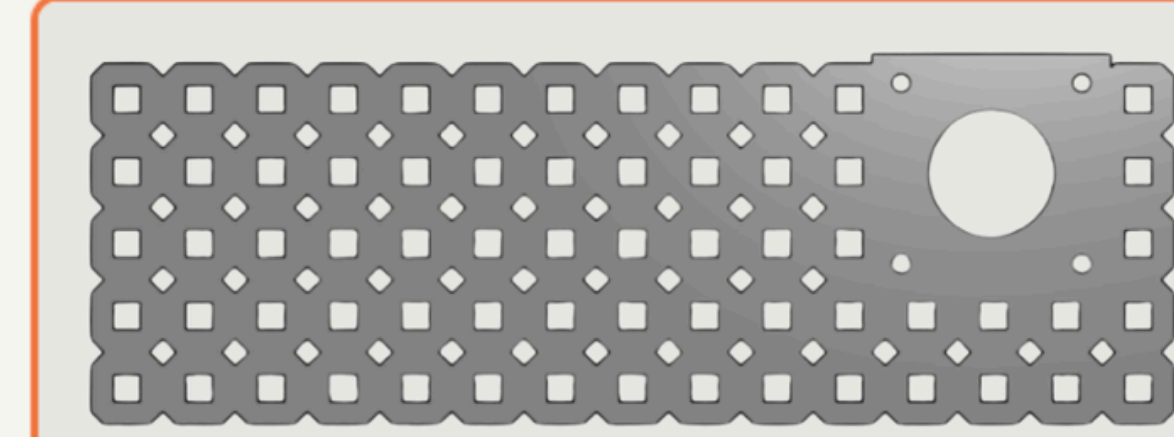
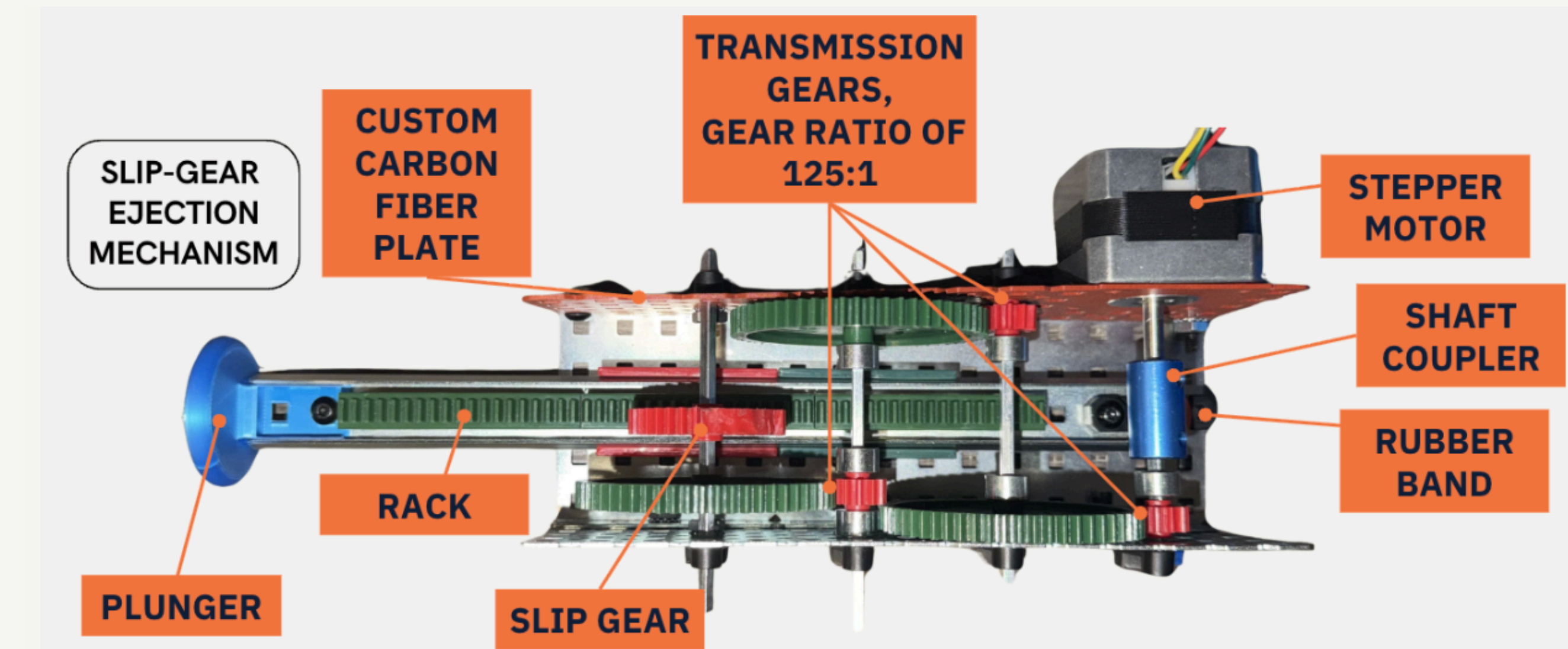
SAFETY INTERLOCK

1 LIMIT SWITCH	2 TRIGGER CONDITIONS	3 SYSTEM RESPONSE	4 ABORT SEQUENCE
- Hatch position sensor - Latch engagement sensor - Redundant signal lines	- Hatch not fully closed - Latch not engaged - Sensor fault / signal loss	- Cycle blocked - OLED alert displayed - Red LED indicator on	- Motor halted immediately - Pressure held / vented - Manual reset required

EJECTION PROCESS

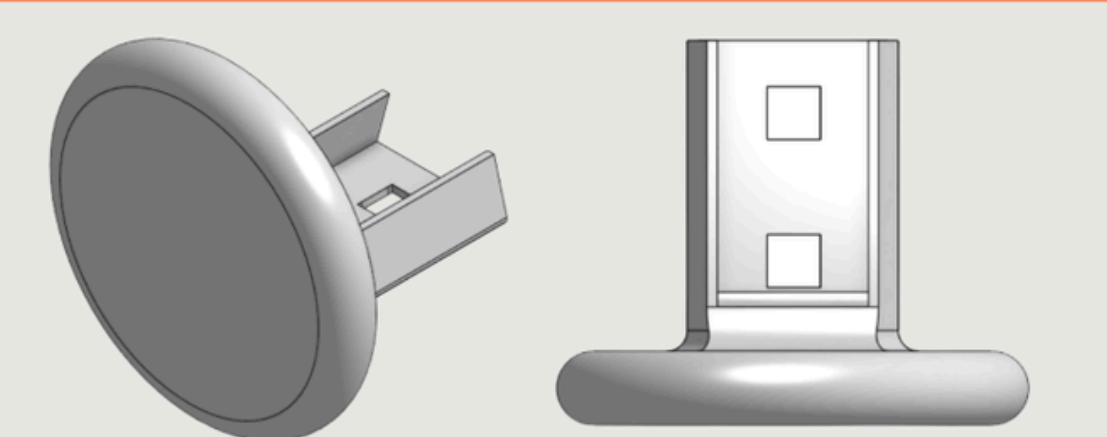


EJECTION MECHANISM



CUSTOM PLATE

Carbon fiber plate that holds the motor in place within the rack and pinion assembly.



PLUNGER

Sits at the end of the rack and pushes the can forward during ejection.

INTERNAL HATCH



LATCH SYSTEM

Dual latches secure the hatch before any cycle step executes.

Replaced barrel bolt



FOLDABLE HANDLE

Folds flat when not in use to minimize space inside the chamber.

PRESSURE ISOLATION

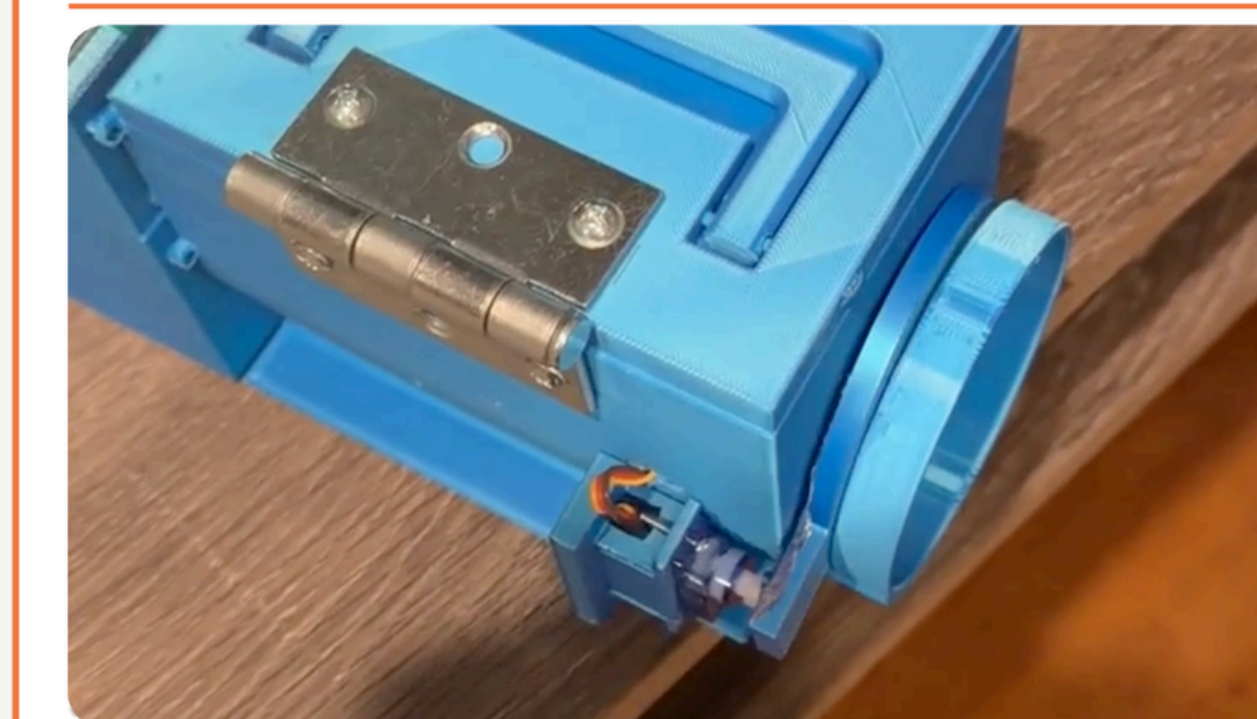
Separates crew cabin from ejection chamber when sealed.

CYCLE INTERLOCK

Limit switch only registers when hatch is fully closed and latched.

EXTERNAL HATCH

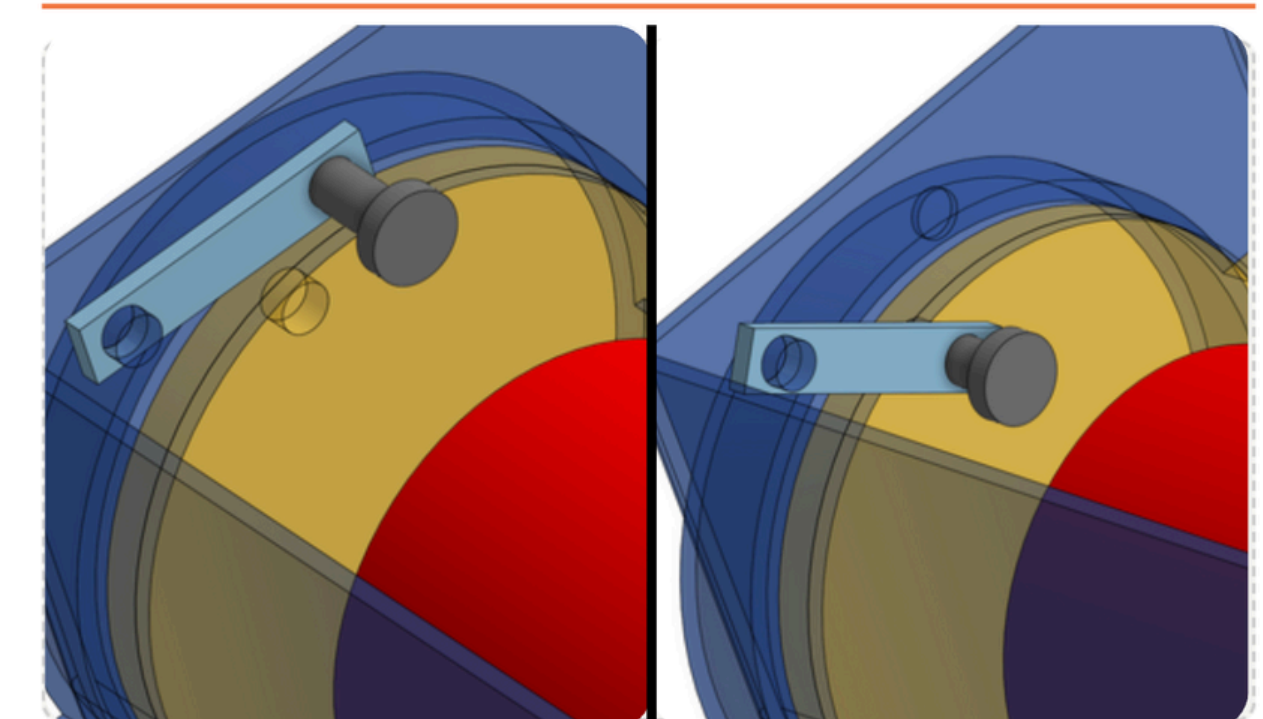
GROOVED SURFACE + LIP



HOW IT WORKS

Servo rotates the door into a groove - the groove forces inward compression as it turns, creating a tight seal.

LOCKING PIN



HOW IT WORKS

A thumb screw drops into a shallow pocket on the hatch face - physically preventing rotation and keeping the door locked shut.